

ALEXIS PARKER

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[LinkedIn](#) • [Portfolio](#)

EDUCATION

CLAREMONT GRADUATE UNIVERSITY | *PhD in Information Systems & Technology* May 2030 (05/30)

- Concentration: Data Science & Analytics

UNIVERSITY OF MICHIGAN SCHOOL OF INFORMATION | *Master of Applied Data Science (MADS)* August 2025 (08/25)

- Projects focused on: Experimental Design, Deep Learning (GANs, RNNs), Bayesian Inference, Health Data Analysis, EDA, Public Health Research, and LLM-based Research Methods

CALIFORNIA STATE UNIVERSITY, SAN BERNARDINO | *Bachelor of Arts in Psychology* June 2019 (06/19)

- Relevant Coursework: Experimental Psychology, Biological Psychology, Drugs and Behavior, Research Methods

SKILLS

Programming & Tools: Python (NumPy, Pandas, scikit-learn, TensorFlow, PyTorch), SQL, R, Git, Jupyter, VS Code, Google Cloud Platform (GCP), Matplotlib

Research Methods: Literature review, qualitative coding, annotation, study ideation, reproducible pipelines

PROFESSIONAL EXPERIENCE

AlphaRose Therapeutics | *Data Science Intern, AI Interfaces* Nov 2025 to Feb 2026

- Developed tokenization and embedding methods for nucleotide sequences, protein sequences, SMILES strings, and PDB files using RDKit, molecular graph structures, and chemoinformatic syntax to support multi-modal, structurally aware modeling in the drug-design pipeline.
- Debugged and refined internal encoding workflows, including sequence encoders and intramolecular torsion calculation scripts, using RDKit, Snakemake, and chemoinformatic tooling to improve reproducibility, transparency, and model readiness.

University of Michigan, School of Information | *Graduate Student Researcher* Jan 2025 to Jan 2026

- Built reproducible data pipelines with Pandas, Seaborn, and scikit-learn; created KDE plots, time series graphs, and regression charts to support exploratory analysis and decision-making.
- Debugged and refined internal encoding workflows, including sequence encoders and torsion calculation scripts, while documenting comparative methods to improve clarity, reproducibility, and integration with downstream machine learning tools.

IHSS Public Authority | *IHSS Caregiver* April 2019 to Present

- Provided long-term, personalized care and coordinated daily support for elderly and disabled clients, maintaining mobility, hygiene, and overall independence.

California State University, San Bernardino | *Behavioral Neuroscience Research Assistant* April to June 2019

- Conducted a behavioral study on substance-related conditioned preference in adolescent rats and analyzed responses using structured protocols and three-way ANOVA.

PROJECT EXPERIENCE

EEG + NHIS Explorer | *Graduate Researcher* June 2025 to August 2025

- Processed EEG recordings (OpenNeuro) with NumPy, SciPy, and MNE-Python; extracted alpha, theta, and beta band power and integrated results with 2024 NHIS sleep survey data (32k respondents)
- Built an interactive Streamlit app with Plotly dashboards (EEG scalp maps, PVT violin plots, national sleep histograms) to connect lab-based and population-level measures

Depression Risk Modeling - NHANES Survey Data | *Graduate Researcher* May 2025 to June 2025

- Merged and cleaned 7 NHANES modules (~12k records) for depression severity prediction; applied KMeans, PCA, and DBSCAN (scikit-learn) to generate features and clusters
- Trained Logistic Regression, Random Forest, and SVM models with hyperparameter tuning; evaluated with SHAP, confusion matrices, and ROC-AUC, engineering socioeconomic predictors from Census data

Depression Index Analysis - CDC BRFSS Survey | *Graduate Researcher* January 2025 to March 2025

- Created a Depression Index using 400,000+ CDC BRFSS responses to model regional variation in self-reported depression symptoms based on PHQ-9 scoring
- Visualized trends in Python, finding Western states scored 15-20% lower than the Midwest and South, supporting research on environmental influences in mental health